

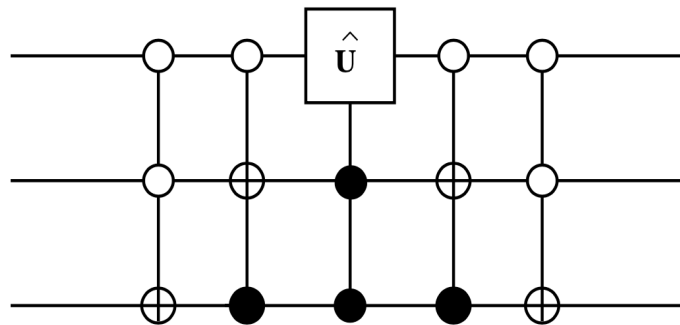
Problem 1. 2-level matrices

Let U be the two level matrix:

$$U = \begin{pmatrix} a & 0 & 0 & 0 & 0 & 0 & 0 & c \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ b & 0 & 0 & 0 & 0 & 0 & 0 & d \end{pmatrix}$$

Then the non-trivial levels correspond to $s = 000$ and $t = 111$.

- i. Check that if we use the Gray code sequence $000 \rightarrow 001 \rightarrow 011 \rightarrow 111$, then we obtain the circuit:



- ii. Check directly that the circuit above actually implements U .
- iii. What is the matrix for the change of basis corresponding to the two controlled gates in the circuit?

Problem 2. QFT

- (a) Work out the quantum Fourier transform matrix F for $n = 3$, and construct the corresponding circuit.
- (b) Show that the inverse of F is given by

$$F^\dagger : |j\rangle \mapsto \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{-2\pi i j k / 2^n} |k\rangle.$$

- (c) Work out the circuit for the inverse of the quantum Fourier transform.
- (d) Show that the square of the QFT applied to $|x\rangle \in \mathbb{C}^{2^n}$ is the multiplication by -1 modulo 2.
- (e) What are the eigenvalue of the QFT?

Hint for point (d):

Recall that the finite series of the form $\sum_{i=0}^{k-1} r^i$ are geometric series of ratio r and that they converge to a value which depends on the ratio, namely:

$$\sum_{i=0}^{k-1} r^i = \begin{cases} \frac{1-r^k}{1-r} & \text{if } r \neq 1 \\ k & \text{if } r = 1 \end{cases}$$

Problem 3.

Let $|\phi\rangle = \sum_{y \in \{0,1\}^n} \beta_y |y\rangle$ be a state such that $\beta_y = 0$ if $s \cdot y = 1 \pmod 2$ and $\beta_y = \frac{1}{\sqrt{2^{n-1}}}$ if $s \cdot y = 0 \pmod 2$, where s is a string different from $00 \dots 0$.

- a) If we perform a Fourier sampling (i.e. the Hadamard transform followed by a measurement in the standard base) on $|\phi\rangle$, what is the probability of obtaining s ?
- b) What is the probability of obtaining a result different from s ?